Lectures On Class Field Theory

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1 Un peu d'histoire

We start with some motivation: this course will be about number fields (i.e. finite extensions $K|\mathbf{Q}$) and their absolute Galois groups $G_K = \mathrm{Gal}(\overline{K}|K)$, where $\overline{K}$ is a fixed algebraic closure of K. Very little is known about the structure of such automorphism groups in general, and a large chunk of modern number theory/arithmetic geometry is devoted to studying them, e.g. via their representations. In some sense, class field theory consists in the study of its 1-dimensional representations; more precisely, it describes the abelianization $G_K^{\mathrm{ab}} = G_K/[G_K, G_K]$ of the absolute Galois group (equivalently, its abelian quotients).

1.1 Reciprocity laws

Let p be a prime number, and let a be an integer coprime to p. The *Legendre symbol* $\left(\frac{a}{p}\right)$ is defined as

$$\left(\frac{a}{p}\right) = \begin{cases} 1 & \text{if } a \text{ is a square modulo } p, \\ -1 & \text{if } a \text{ is not a square modulo } p. \end{cases}$$

Euler proved that $\left(\frac{a}{p}\right) = a^{\frac{p-1}{2}} \bmod p$, and **Gauss** proved the following law:

Theorem 1.1 (Quadratic reciprocity)**.** Let $a \neq b$ be different odd primes. Then,

$$\left(\frac{a}{b}\right) \cdot \left(\frac{b}{a}\right) = (-1)^{\frac{a-1}{2}\frac{b-1}{2}}.$$

Morever, the following subsidiary laws hold:

1. $\left(\frac{-1}{p}\right) = (-1)^{\frac{p-1}{2}}$;
2. $\left(\frac{2}{p}\right) = (-1)^{\frac{p^2-1}{8}}$.

These laws, together with its multiplicativity in the parameter a, make it possible to compute the Legendre symbol.

Another way of thinking about class field theory is as an attempt to define a "very general" reciprocity law, from which the quadratic reciprocity law and its higher degree analogues can be deduced. One such generalization is **Hilbert**'s reciprocity law: recall that $\mathbf{Q}$ embeds into the p-adic field $\mathbf{Q}_p$ for every prime p, and also into the real numbers $\mathbf{Q}_\infty \coloneqq \mathbf{R}$. For every prime p (including $p = \infty$) and for all $a, b \in \mathbf{Q}^\times$, we define the *Hilbert symbol* $(a, b)_p$ as

$$(a, b)_p = \begin{cases} 1 & \text{if } ax^2 + by^2 = 1 \text{ has a solution in } \mathbf{Q}_p^2, \\ -1 & \text{otherwise.} \end{cases} \quad asd$$

Recall that the norm $N : \mathbf{Q}_p(\sqrt{a}) \to \mathbf{Q}_p$ is defined as $N(x + y\sqrt{a}) = x^2 - ay^2$ if $a \notin \mathbf{Q}_p^{\times 2}$, and as $N = \mathrm{id}$ otherwise. More generally, if $L|K$ is a finite Galois extension with group G, the norm $N_{L|K} : L \to K$ is

$$N(x) = \prod_{\sigma \in G} \sigma(x)$$

Lemma 1.2. The following statements are equivalent:

1. $(a,b)_p = 1$;
2. The quadratic equation $z^2 = ax^2 + by^2$ has a non-trivial solution;
3. $a \in \mathbf{Q}_p$ is the norm of an element of $\mathbf{Q}_p(\sqrt{b})$.
4. $b \in \mathbf{Q}_p$ is the norm of an element of $\mathbf{Q}_p(\sqrt{a})$.

Proof. The equivalence $1 \iff 2$ is by definition; and $3 \iff 4$ is clear. For $2 \iff 3$, if $b \in \mathbf{Q}_p^{\times 2}$, then we are done. Otherwise, if $b \notin \mathbf{Q}_p^{\times 2}$, then $N(\alpha + \beta\sqrt{b}) = \alpha^2 - b\beta^2$, and if $a \coloneqq \alpha^2 - b\beta^2$ we see that $z^2 = ax^2 + by^2$ has a non-trivial solution, and vice versa. □

For $p \neq 2, \infty$ and $a, b \in \mathbf{Z}$, write $a = p^{v_p(a)}u$ and $b = p^{v_p(b)}v$ with $p \nmid uv$. The relation between the Hilbert symbol and the Legendre symbol is

$$(a,b)_p = (-1)^{\frac{p-1}{2}v_p(a)v_p(b)} \left(\frac{u}{p}\right)^{v_p(b)} \left(\frac{v}{p}\right)^{v_p(a)} \tag{1.1}$$

$$= \overline{(-1)^{v_p(a)v_p(b)} a^{v_p(b)} b^{-v_p(a)}}^{(p-1)/2}. \tag{1.2}$$

where the bar denotes reduction modulo p. In particular, if p does not divide a, then $(a,b)_p = \left(\frac{a}{p}\right)^{v_p(b)}$ coincides with the Legendre symbol.

Theorem 1.3 (Hilbert reciprocity law)**.** For all $a, b \in \mathbf{Q}^\times$, we have

$$\prod_p (a,b)_p = 1$$

As for computation, one shows that if p does not divide $2ab$ (including $p = \infty$ for $a, b > 0$), then $(a,b)_p = 1$. Moreover, $(a,b)_2 = (-1)^{\frac{a-1}{2}\frac{b-1}{2}}$ for odd a, b. From this and multiplicativity in the first variable, we immediately deduce the quadratic reciprocity law by taking a, b to be odd primes.

An advantage of this formulation is that it generalizes to number fields K, by considering the completions of K at all of its (finite and infinite) "places". Hilbert then went on to pose his 9th problem, in which he asked for a generalization of this law to n-th powers in arbitrary number fields; this is what class field theory achieves. We first consider the case of extensions which contain a primitive n-th root of unity:

1.2 Kummer case

Assume now $K|\mathbf{Q}$ is a finite extension such that $\mu_n \subset K$, and let $\mathcal{O}_K$ be its ring of integers. If $P \subset \mathcal{O}_K$ is a prime ideal, we would like to define a symbol $(a,b)_P \in \mu_n$ such that a product formula holds, i.e.

$$\prod_P (a,b)_P = 1$$

if $a, b \in K^\times$. Moreover, for P finite (which is always the case for $n \neq 2$) which does not divide (n) we shall have

$$(a,b)_P = \overline{(-1)^{v_P(a)v_P(b)} a^{v_P(b)} b^{-v_P(a)}}^{(q-1)/n},$$

where $q = \#(\mathcal{O}_K/P)$. In particular, if P does not divide (a), then $(a,b)_P = a^{\frac{q-1}{n} v_P(b)} \bmod P$.

So, if $v_P(b) = 1$, then

$$\left(\frac{a}{P}\right) \coloneqq (a,b)_P$$

does not depend on the choice of b and it is the unique n-th root of unity, that is $a^{\frac{q-1}{n}} \bmod P$; in other words, $\left(\frac{a}{P}\right) = 1$ if and only if $x^n = a \bmod P$ has a solution. Some more properties of this symbol are:

1. $(a,b)_P = 1 \iff b$ is a norm from $K_P(\sqrt[n]{a}) \iff a$ is a norm from $K_P(\sqrt[n]{b})$;
2. $(a,b)_P$ is multiplicative in both variables and anticommutative;

When μ_n is *not* contained in K_P, an "Hilbert" symbol still exists but is longer associated to $a, b \in K^\times$. Suppose for a moment that $\mu_n \subset K_P$. Then, the extension $K_P(\sqrt[n]{a})|K_P$ is Galois for all $a \notin K_P^{\times n}$, and its Galois group is

$$G_a \coloneqq \mathrm{Gal}(K_P(\sqrt[n]{a})|K_P) \cong \mathbf{Z}/n\mathbf{Z}$$

Then, the group $\mathrm{Hom}(G_a, \mu_n) \cong \mathbf{Z}/n\mathbf{Z}$ is generated by the character

$$\overline{\chi_a} : \sigma \mapsto \frac{\sigma(\sqrt[n]{a})}{\sqrt[n]{a}} \in \mu_n \cong \mathbf{Z}/n\mathbf{Z}.$$

In general, we shall construct a generalized Hilbert symbol

$$\begin{aligned} \mathrm{Hom}\left(\mathrm{Gal}(\overline{K_P}|K_P), \mathbf{Z}/n\mathbf{Z}\right) \times K_P^\times &\to \mathbf{Z}/n\mathbf{Z}, \\ [\chi, b] &\mapsto (\chi, b)_P \end{aligned}$$

and we'll thus be able to prove the **Artin** reciprocity law: fix some $b \in K^\times$, and suppose that χ is a character in $\mathrm{Hom}(G_K, \mathbf{Z}/n\mathbf{Z})$. Then

$$\sum_P (\chi_P, b) = 0,$$

where χ_P is the restriction of χ to $G_{K_P}^{\mathrm{ab}}$ (recall that it embeds into G_K^{ab}).

The generalized Hilbert symbol defines a reciprocity map

$$K_P^\times \to G_{K_P}^{\mathrm{ab}},$$

which is "almost" an isomorphism. In particular, for every n we'll have an isomorphism of finite groups

$$K_P^\times/n \to G_{K_P}^{\mathrm{ab}}/n.$$

This is one of the remarkable features of class field theory, giving a description of the abelian extensions of a field in terms of its arithmetic. To define such a map, it's enough to define the dual map

$$\begin{aligned} \mathrm{Hom}(G_{K_P}^{\mathrm{ab}}/n, \mathbf{Z}/n\mathbf{Z}) &\to \mathrm{Hom}(K_P^\times/n, \mathbf{Z}/n\mathbf{Z}), \\ [\chi] &\mapsto [b \mapsto (\chi, b)_P]. \end{aligned}$$

2 Computing the Hilbert symbol

Recall the definition: for $a, b \in \mathbf{Q}_p^\times$,

$$(a,b)_p = \begin{cases} 1 & \text{if } ax^2 + by^2 = 1 \text{ is solvable in } \mathbf{Q}_p, \\ -1 & \text{otherwise.} \end{cases}$$

We have seen that

$(a,b)_p = 1 \iff z^2 - ax^2 - by^2$ has a non-trivial solution in $\mathbf{Q}_p^3 \iff b$ is a norm from $\mathbf{Q}_p(\sqrt{a})$,

where $N(x + y\sqrt{a}) = x^2 - ay^2$.

Theorem 2.1 (Hilbert reciprocity law)**.** For all $a, b \in \mathbf{Q}^\times$,

$$\prod_p (a,b)_p = 1,$$

where the product runs over all primes p, including $p = \infty$.

Note that the product is finite: $(a,b)_p = 1$ for all $p \nmid 2ab$ (see Lemma 2.4 below).

Lemma 2.2. Let $a, b, c \in \mathbf{Q}_p^\times$. Then

1. $(1,c)_p = 1$;
2. $(-c,c)_p = 1$;
3. if $(a,c)_p = 1$, then $(ab,c)_p = (a,c)_p(b,c)_p$ (this holds in general, see the Corollary below);
4. $(c,c)_p = (-1,c)_p$.

Proof. 1. Consider $N : \mathbf{Q}_p(\sqrt{c}) \to \mathbf{Q}_p$. Then $N(1) = 1$.

2. If $c \in \mathbf{Q}_p^{\times 2}$, then $\mathbf{Q}_p(\sqrt{c}) = \mathbf{Q}_p$, the norm is the identity and $N(-c) = -c$. If $c \notin \mathbf{Q}_p^{\times 2}$, then $N(\sqrt{c}) = -c$. In both cases $-c$ is a norm from $\mathbf{Q}_p(\sqrt{c})$.

3. If $a = N(x)$ and $b = N(y)$, then $ab = N(x)N(y) = N(xy)$, since the norm is multiplicative by definition.

   Moreover, if $a = N(x)$, then $a^{-1} = N(x^{-1})$; if in addition $ab = N(z)$, then

   $$b = a^{-1}(ab) = N(x^{-1})N(z) = N(x^{-1}z).$$

   So, if $(a,c)_p = 1$ (i.e. $a = N(x)$ is a norm from $\mathbf{Q}_p(\sqrt{c})$), then $(b,c)_p = 1 \iff (ab,c)_p = 1$.

4. By (2) and (3),

   $$(c,c)_p = (-c,c)_p(c,c)_p = (-c^2,c)_p = (-1,c)_p,$$

   because in general $(ad^2,b)_p = (a,b)_p$. □

Remark 2.3. Since $(ad^2, b)_p = (a, b)_p$, the Hilbert symbol is a well-defined map

$$(\,\cdot\,,\cdot\,)_p : \left(\mathbf{Q}_p^\times/\mathbf{Q}_p^{\times 2}\right) \times \left(\mathbf{Q}_p^\times/\mathbf{Q}_p^{\times 2}\right) \to \mu_2.$$

Lemma 2.4. Let $p > 2$. Then $(u, v)_p = 1$ for all $u, v \in \mathbf{Z}_p^\times$. (Recall that $\mathbf{Z}_p$ is local and $\mathbf{Z}_p^\times = \mathbf{Z}_p \setminus (p)$.)

Proof. Let $\overline{u} \coloneqq u \bmod p$ and $\overline{v} \coloneqq v \bmod p$; they are non-zero elements of $\mathbf{F}_p$ (units of the local ring). We first show that $1 - \overline{u}x^2 - \overline{v}y^2$ has a solution in $\mathbf{F}_p$. Since $\mathbf{F}_p^\times$ is cyclic of order $p-1$,

$$\#\{x^2 : x \in \mathbf{F}_p\} = \frac{p+1}{2}.$$

Similarly, if $S_{\overline{u}} \coloneqq \{\overline{u}x^2 : x \in \mathbf{F}_p\}$ and $S_{\overline{v}} \coloneqq \{1 - \overline{v}y^2 : y \in \mathbf{F}_p\}$, then $\#S_{\overline{u}} = \#S_{\overline{v}} = \frac{p+1}{2}$. Thus, $S_{\overline{u}} \cap S_{\overline{v}} \neq \emptyset$ by the

We conclude by Hensel's lemma: □

Lemma 2.5 (Hensel)**.** Let $f \in \mathbf{Z}_p[X]$ and $a_0 \in \mathbf{Z}_p$ with $f(a_0) \equiv 0 \mod p$ and $f'(a_0) \not\equiv 0 \mod p$. Then there exists $a \in \mathbf{Z}_p$ with $f(a) = 0$ and $a \equiv a_0 \mod p$.

This allows us to lift a solution mod p to a solution in $\mathbf{Z}_p$ (to apply it in the proof above, we have to fix one variable).

Proposition 2.6. Let $p > 2$ and $a, b \in \mathbf{Q}_p^\times$, with $a = p^\alpha u$, $b = p^\beta v$ and $u, v \in \mathbf{Z}_p^\times$. Then

$$(a, b)_p = (-1)^{\alpha\beta\frac{p-1}{2}} \left(\frac{\overline{u}}{p}\right)^{-\beta} \left(\frac{\overline{v}}{p}\right)^{\alpha}.$$

Proof. Since $(a, b)_p$ depends only on $a, b \bmod \mathbf{Q}_p^{\times 2}$, we may assume $\alpha, \beta \in \{0, 1\}$.

- $\alpha = \beta = 0$: this is Lemma 2.4.
- $\alpha = 1$, $\beta = 0$: we need $(pu, v)_p = \left(\frac{\overline{v}}{p}\right)$. By Lemma 2.4 we have $(u^{-1}, v)_p = 1$, hence by Lemma 2.2(3)
$$(pu, v)_p = (pu, v)_p \underbrace{(u^{-1}, v)_p}_{=1} = (p, v)_p.$$
If $v \in \mathbf{Q}_p^{\times 2}$, then $(p, v)_p = (p, 1)_p = 1 = \left(\frac{\overline{v}}{p}\right)$ (note that $\overline{v}$ is then also a square mod p). If $v \notin \mathbf{Q}_p^{\times 2}$, then $(p, v)_p = -1$, because $z^2 - px^2 - vy^2$ has no non-trivial solution mod p, hence none in $\mathbf{Q}_p$ by homogeneity; also $\left(\frac{\overline{v}}{p}\right) = -1$ (by Hensel, since $v \notin \mathbf{Q}_p^{\times 2}$).

- $\alpha = \beta = 1$: we need $(pu, pv)_p = (-1)^{\frac{p-1}{2}} \left(\frac{\overline{u}}{p}\right)\left(\frac{\overline{v}}{p}\right)$. By Lemma 2.2(2) we have $(-pv, pv)_p = 1$, so by Lemma 2.2(3)

$$(pu, pv)_p = (pu, pv)_p \underbrace{(-pv, pv)_p}_{=1} = (-p^2uv, pv)_p = (-uv, pv)_p = (pv, -uv)_p,$$

and now we are in the previous case, which gives

$$(pv, -uv)_p = \left(\frac{\overline{-uv}}{p}\right) = \left(\frac{-1}{p}\right)\left(\frac{\overline{u}}{p}\right)\left(\frac{\overline{v}}{p}\right),$$

where $\left(\frac{-1}{p}\right) = (-1)^{\frac{p-1}{2}}$, and we used that the Legendre symbol is multiplicative. □

Corollary 2.7. Let $p > 2$. Then $(a, c)_p(b, c)_p = (ab, c)_p$ for all $a, b, c \in \mathbf{Q}_p^\times$. In particular, $(\,\cdot\,,\cdot\,)_p$ is bi-multiplicative.

Proof. Write $a = p^\alpha u$, $b = p^\beta v$ and $c = p^\gamma w$ with $u, v, w \in \mathbf{Z}_p^\times$. By Proposition 2.6,

$$\begin{aligned}(a, c)_p(b, c)_p &= (-1)^{\frac{p-1}{2}\alpha\gamma}\left(\frac{\overline{u}}{p}\right)^\gamma\left(\frac{\overline{w}}{p}\right)^\alpha \cdot (-1)^{\frac{p-1}{2}\beta\gamma}\left(\frac{\overline{v}}{p}\right)^\gamma\left(\frac{\overline{w}}{p}\right)^\beta \\ &= (-1)^{\frac{p-1}{2}\gamma(\alpha+\beta)}\left(\frac{\overline{uv}}{p}\right)^\gamma\left(\frac{\overline{w}}{p}\right)^{\alpha+\beta} = (ab, c)_p.\end{aligned}$$

□

Case $p = 2$. If $a = 2^\alpha u$ and $b = 2^\beta v$ with $u, v \in \mathbf{Z}_2^\times$, define

$$\varepsilon(u) \coloneqq \frac{u-1}{2} \bmod 2, \qquad \omega(u) \coloneqq \frac{u^2-1}{8} \bmod 2.$$

Then

$$(a, b)_2 = (-1)^{\varepsilon(u)\varepsilon(v)+\alpha\omega(v)+\beta\omega(u)}.$$

(Proof not given.) In particular, $(\,\cdot\,,\cdot\,)_p$ is bi-multiplicative for every p.

Proof of Theorem 2.1. Using bi-multiplicativity, we reduce to the case where each of a, b is either -1 or a prime.

- Case $a = -1$: homework.
- Case $a = b = q$ prime: $(q, q)_p = (-1, q)_p$ by Lemma 2.2(4), so we are reduced to the previous case.
- Case $a = 2$, $b > 2$ prime: $(2, b)_p = 1$ for $p \notin \{2, b\}$ (also for $p = \infty$, as $2, b > 0$). Moreover, by the $p = 2$ formula,

$$(2, b)_2 = (-1)^{\omega(b)},$$

and by Proposition 2.6

$$(2, b)_b = \left(\frac{2}{b}\right) = (-1)^{\frac{b^2-1}{8}}.$$

The product of these two factors is 1.

- Case $a \neq b$ odd primes: $(a,b)_p = 1$ for $p \notin \{2, a, b\}$ (as well as $p = \infty$), and

$$(a,b)_p = \begin{cases} (-1)^{\frac{a-1}{2}\frac{b-1}{2}} & p = 2, \\ \left(\frac{a}{b}\right) & p = b, \\ \left(\frac{b}{a}\right) & p = a. \end{cases}$$

  Hence the product is 1 by quadratic reciprocity. □

3 Cohomology of groups

Already for the local part of theory we'll require some input from homological algebra, mainly the description of the first cohomology group $H^1(G, A)$ for G some Galois group, and its interpretation in terms of crossed homomorphisms. A concise reference for this material can be found in the notes [2]. We recall some results here for the reader's convenience:

Theorem 3.1 (Cohomology of cyclic groups)**.** Let G be a finite cyclic group of order n, generated by σ. Then,

$$\begin{cases} H^0(G, A) = A^G, \\ H^1(G, A) = A/(1-\sigma)A, \\ H^2(G, A) = A^G/NA, \\ H^i(G, A) \cong H^{i+2}(G, A) \text{ for all } i \geq 1, \end{cases}$$

This comes out of the "periodic" resolution of $\mathbf{Z}$ as a trivial G-module:

$$\ldots \xrightarrow{\sigma-1} \mathbf{Z}[G] \xrightarrow{N} \mathbf{Z}[G] \xrightarrow{\sigma-1} \mathbf{Z}[G] \xrightarrow{N} \mathbf{Z}[G] \xrightarrow{\varepsilon} \mathbf{Z} \to 0,$$

where $N = 1 + \sigma + \ldots + \sigma^{n-1}$ is the norm map and ε sends σ to 1. A similar computation holds for the infinite cyclic group $G = \mathbf{Z}$:

$$H^0(\mathbf{Z}, A) = A^{\mathbf{Z}}, \qquad H^1(\mathbf{Z}, A) = A/(1-\sigma)A, \qquad H^i(\mathbf{Z}, A) = 0 \text{ for all } i \geq 2.$$

For a subgroup $H \subset G$, we denote by

$$M_H^G(A) \coloneqq \mathrm{Hom}_H(\mathbf{Z}[G], A)$$

the G-module with action

$$\sigma(\phi)(g) = \phi(\sigma^{-1}g), \qquad \sigma \in G, g \in G, \phi \in M_H^G(A).$$

Then, *Shapiro*'s lemma tells us that there is a canonical isomorphism

$$H^i(G, M_H^G(A)) \cong H^i(H, A). \tag{3.1}$$

3.1 Restriction, Inflation and Corestriction

Let H be a subgroup of G. The *restriction* map is defined on cocycles as

$$\mathrm{Res} : H^i(G, A) \to H^i(H, A), \quad \mathrm{Res}([f]) = [f|_H],$$

which explains the name: $f|_H$ is just restriction to H.

For a normal subgroup $H \triangleleft G$, the quotient G/H acts on A^H and we have the *inflation* map

$$\mathrm{Inf} : H^i(G/H, A^H) \to H^i(G, A), \quad \mathrm{Inf}([f]) = [f \circ \pi],$$

where $\pi : G \to G/H$ is the canonical projection.

The *corestriction* map goes in the opposite direction of restriction, and is defined as follows: let $H \subset G$ be a subgroup of finite index This is well-defined and gives a map

$$\mathrm{Cor} : H^i(H, A) \to H^i(G, A).$$

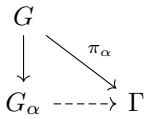
3.2 Profinite groups

Let $G = \varprojlim G_\alpha$ be a profinite group (that is, an inverse limit of finite groups). This is a topological group, whose topology is induced by the discrete topology on the finite groups G_α.

We have natural continuous maps $\pi_\alpha : G \to G_\alpha$, and their kernels U_α form a system of open neighborhoods of $1 \in G$.

Remark 3.2.

- We may assume that the G_α coincide with **all** finite quotients of G by open normal subgroups, as if Γ is such a quotient, then there is some α such that U_α has trivial image in Γ: indeed, Γ is finite and the intersection of all U_α is trivial. So, we get a factorization



- The inverse limit of a system of finite groups is compact and totally disconnected. With compact here we also include the Hausdorff property, which holds by Tychonoff's theorem. Moreover, the open subgroups of G are exactly the closed subgroups of finite index.

3.3 Cohomology of profinite groups

Let $G = \varprojlim G_\alpha$ be a profinite group, and let A be a *continuous* G-module, i.e. an abelian group with an action of G by automorphisms such that $\forall a \in A$, the stabilizer G_a is open in G (exercise: this is equivalent to the map $G \times A \to A$ being continuous, where A has the discrete topology). In particular, if $G_\alpha = G/U_\alpha$, then A^{U_α} is a G_α-module, and we define:

Definition 3.3.

$$H^i_{\mathrm{cont}}(G, A) \coloneqq \varinjlim H^i(G_\alpha, A^{U_\alpha}),$$

where the limit is taken with respect to the inflation maps induced by the projections $G_\beta \twoheadrightarrow G_\alpha$:

$$H^i(G_\alpha, A^{U_\alpha}) \to H^i(G_\beta, A^{U_\beta}).$$

We'll immediately drop the subscript cont, writing $H^i(G, A)$.

In the case where $G = \mathrm{Gal}(k^s|k)$ is the absolute Galois group of a field k, we'll use the notation $H^i(k, A)$ for $H^i(G, A)$.

Proposition 3.4. If G is profinite and A is a continuous G-module, then $H^i(G, A)$ is a torsion abelian group for all $i > 0$.

Proof. The groups $H^i(G_\alpha, A^{U_\alpha})$ are torsion: they are killed by the order of G_α. Therefore, their direct limit $H^i(G, A)$ is also torsion. □

Corollary 3.5. If moreover A is a $\mathbf{Q}$-vector space, then $H^i(G, A) = 0$ for all $i > 0$.

Remark 3.6. Given a finite group G and a finitely generated G-module A, the cohomology groups $H^i(G, A)$ are finite for all $i > 0$. This follows from the description of cohomology in terms of the standard resolution and the torsion property above.

Lemma 3.7. Let G be profinite with (U_α) a system of open normal subgroups. Then,

- If H is a closed subgroup of G, there is a canonical isomorphism
$$H^i(H, A) \cong \varinjlim H^i(H/(H \cap U_\alpha), A^{H \cap U_\alpha})$$
which shows that H is profinite;
- If moreover H is normal in G, then $G/H \cong \varprojlim G/(HU_\alpha)$ is profinite.

Therefore, we can define restriction, corestriction and inflation maps for profinite groups by taking the direct limit of the corresponding maps for finite groups: for instance, restriction is induced by the maps

$$H^i(G/U_\alpha, A^{U_\alpha}) \xrightarrow{\mathrm{res}} H^i(H/(H \cap U_\alpha), A^{H \cap U_\alpha}).$$

Since we defined cohomology explicitly as a direct limit, we have to show "by hand" the existence of the long exact sequence in cohomology associated with a short exact sequence of G-modules $0 \to A \to B \to C \to 0$.

Lemma 3.8. Let $(A_\alpha, \phi_{\alpha\beta})$, $(B_\alpha, \psi_{\alpha\beta})$ and $(C_\alpha, \rho_{\alpha\beta})$ be direct systems of G_α-modules, and assume given short exact sequences

$$0 \to A_\alpha \xrightarrow{f_\alpha} B_\alpha \xrightarrow{g_\alpha} C_\alpha \to 0$$

for all α which are compatible with the transition maps (i.e. morphisms of direct systems). Then, the direct limit of these sequences is exact.

The non-trivial part is defining the connecting homomorphism

$$\delta : H^i(G, C) \to H^{i+1}(G, A).$$

Let $U \subset G$ be an open normal subgroup. If $K_U \coloneqq \operatorname{coker}(B^U \to C^U,)$ then $\varinjlim_U K_U = 0$. By the lemma, we have a short exact sequence

$$\varinjlim H^i(G/U, (B^U/A^U)) \to \varinjlim H^i(G/U, C^U) \to \varinjlim H^{i+1}(G/U, K_U)$$

and the last term is 0 by the above. Therefore, for small enough U we can lift a cocycle $f \in Z^i(G, C)$ to some element in $\varinjlim H^i(G/U, (B^U/A^U))$, and then apply the connecting homomorphism coming from the short exact sequence $0 \to A^U \to B^U \to B^U/A^U \to 0$ to get an element in $\varinjlim H^{i+1}(G/U, A^U)$, which has a map into $H^{i+1}(G, A)$ (of course, one should check that this does not depend on any of the choices made).

3.4 Kummer Theory

Proposition 3.9 (Hilbert's Theorem 90)**.** Let $K|k$ be a finite Galois extension with group $G = \operatorname{Gal}(K|k)$. Then

$$H^1(G, K^\times) = 0.$$

Consequently, by passing to the limit over all such extensions we get

$$H^1(k, (k^s)^\times) = 0.$$

Lemma 3.10 (Dedekind)**.** If $\sigma_1, \ldots, \sigma_n$ are the elements of G, than they are linearly independent over K.

Proof. Suppose they are linearly dependent, and let

$$a_1\sigma_1 + \ldots + a_n\sigma_n = 0$$

be the *shortest* nontrivial linear relation. Then, choosing $x \in K$ such that $\sigma_1(x) \neq \sigma_2(x)$ and $y \in K$ is arbitrary, then

$$\begin{aligned} 0 &= a_1\sigma_1(xy) + \ldots + a_n\sigma_n(xy) \\ &= a_1\sigma_1(x)\sigma_1(y) + \ldots + a_n\sigma_n(x)\sigma_n(y) \\ &= \sigma_1(x)(a_1\sigma_1(y) + \ldots + a_n\sigma_n(y)) + (\sigma_2(x) - \sigma_1(x))(a_2\sigma_2(y) + \ldots + a_n\sigma_n(y)). \end{aligned}$$

Putting this together with the original relation evaluated at y and multiplied by $\sigma_1(x)$, we get a shorter relation

$$(\sigma_2(x) - \sigma_1(x))(a_2\sigma_2(y) + \ldots + a_n\sigma_n(y)) = 0,$$

which is a contradiction. □

Proof of Proposition. Let $\varphi : G \to K^\times$ be a 1-cocycle, with the cocycle condition being $\varphi(\sigma\tau) = \varphi(\sigma) \cdot \sigma(\varphi(\tau))$. We want to show that φ is a coboundary, i.e. there exists some $y \in K^\times$ such that $\varphi(\tau) = \tau(y) \cdot y^{-1}$ for all $\tau \in G$. Consider the element

$$\Phi = \sum_{\sigma \in G} \varphi(\sigma) \cdot \sigma \in K[G].$$

By the lemma, Φ is non-zero. Therefore there exist $x, y \in K^\times$ such that

$$y = \Phi(x) = \sum_{\sigma \in G} \varphi(\sigma) \cdot \sigma(x),$$

and applying τ to both sides gives

$$\tau(y) = \sum_{\sigma \in G} \tau\varphi(\sigma) \cdot \tau\sigma(x) = \sum_{\sigma \in G} \frac{\varphi(\tau\sigma)}{\varphi(\tau)} \tau\sigma(x) = \frac{1}{\varphi(\tau)} \sum_{\sigma \in G} \varphi(\tau\sigma) \cdot \tau\sigma(x) = \frac{1}{\varphi(\tau)} \sum_{\sigma \in G} \Phi(x) = \frac{y}{\varphi(\tau)},$$

□

The main result of this section is the following:

Proposition 3.11. Let k be a field and m a positive integer which is prime to $\mathrm{char}(k)$ (this implies that $\mu_m \subset k^s$). Then, there is a canonical isomorphism

$$H^1(k, \mu_m) \cong k^\times / k^{\times m}.$$

whose inverse sends the class of $a \in k^\times / k^{\times m}$ to the class of the cocycle $\sigma \mapsto \sigma(\alpha) \cdot \alpha^{-1}$, where $\alpha \in (k^s)^\times$ is any m-th root of a.

Proof. Consider the short exact sequence of G_k-modules

$$1 \to \mu_m \to (k^s)^\times \xrightarrow{(\cdot)^m} (k^s)^\times \to 1.$$

By Hilbert's Theorem 90, we have $H^1(k, (k^s)^\times) = 0$. Therefore, the long exact sequence in cohomology gives

$$H^0(k, (k^s)^\times) \xrightarrow{(\cdot)^m} H^0(k, (k^s)^\times) \to H^1(k, \mu_m) \to 0,$$

and the cokernel of the first map is exactly $k^\times / k^{\times m}$, since $H^0(k, (k^s)^\times) = (k^s)^{\times G_k} = k^\times$. The last statement then follows by writing out the connecting map explicitly (lifting corresponds to taking an m-th root…). □

Example 3.12. If moreover $\mu_m \subset k$, then $H^1(k, \mu_m) = H^1(k, \mathbf{Z}/m\mathbf{Z}) = \mathrm{Hom}(G_k, \mathbf{Z}/m\mathbf{Z})$ (the action is trivial). In this case, we deduce that every finite extension $K|k$ with $\mathrm{Gal}(K|k) \cong \mathbf{Z}/m\mathbf{Z}$ is of the form $K = k(\alpha)$ with $\alpha^m \in k$.

3.5 Cup-products

Let G be a group and A, B be G-modules. The module $A \otimes_{\mathbf{Z}} B$ is a G-module with the diagonal action $\sigma(a \otimes b) = \sigma(a) \otimes \sigma(b)$. The cup-product is a bilinear pairing

$$\cup : H^i(G, A) \times H^j(G, B) \to H^{i+j}(G, A \otimes B)$$
$$(a, b) \mapsto a \cup b$$

which is functorial in A, B and graded-commutative: $a \cup b = (-1)^{ij} b \cup a$; there will be a version for profinite groups by passing to the limit. Let us sketch the application of this gadget to the Hilbert symbol:

Definition 3.13. For a field k, the *Brauer group* of k is

$$\mathrm{Br}(k) \coloneqq H^2(k, (k^s)^\times).$$

A non-trivial fact is that elements of $\mathrm{Br}(k)$ correspond to central division algebras which are finite-dimensional over k. Then, another piece of the Kummer long exact sequence is

$$H^1(k, (k^s)^\times) = 0 \to H^2(k, \mu_n) \xrightarrow{(\cdot)^n} H^2(k, (k^s)^\times),$$

from which we deduce that $H^2(k, \mu_n) \cong {}_n\mathrm{Br}(k)$ is precisely the n-torsion of the Brauer group.

Suppose that $\mu_n \subset k$, and take $a, b \in k^\times$. Then, we can take their classes in $k^\times / k^{\times n} \cong H^1(k, \mu_n) = H^1(k, \mathbf{Z}/n\mathbf{Z})$, and take their cup-product:

$$H^1(k, \mathbf{Z}/n\mathbf{Z}) \times H^1(k, \mathbf{Z}/n\mathbf{Z}) \to H^2(k, \mathbf{Z}/n\mathbf{Z} \otimes_{\mathbf{Z}} \mathbf{Z}/n\mathbf{Z}) \cong {}_n\mathrm{Br}(k).$$

Theorem 3.14 (Hasse)**.** Let $k|\mathbf{Q}_p$ be a finite extension. Then,

$$\mathrm{Br}(k) \cong \mathbf{Q}/\mathbf{Z}.$$

In particular we shall deduce that, for any n, ${}_n\mathrm{Br}(k) \cong \frac{1}{n}\mathbf{Z}/\mathbf{Z}$. Putting these together, we get a map

$$(\cdot, \cdot)_n : k^\times / k^{\times n} \times k^\times / k^{\times n} \to H^1(k, \mathbf{Z}/n\mathbf{Z}) \times H^1(k, \mathbf{Z}/n\mathbf{Z}) \to {}_n\mathrm{Br}(k) \cong \mathbf{Z}/n\mathbf{Z},$$

which is the *Hilbert symbol*. What happens if $\mu_n \not\subset k$? Given a character $\chi : G_k \to \mathbf{Z}/n\mathbf{Z}$, this lives in the first cohomology group $H^1(k, \mathbf{Z}/n\mathbf{Z})$, so we can take its cup-product with the class of any $b \in k^\times$ inside $H^1(k, \mu_n)$:

$$(\chi, b) \coloneqq \chi \cup b \in H^2(k, \mu_n) \overset{\text{Hasse}}{\cong} \mathbf{Z}/n\mathbf{Z}.$$

Construction 3.15. Let $A^\bullet, B^\bullet$ be complexes of e.g. abelian groups. The tensor product of

these complexes is defined as follows: consider the double complex

$$\begin{array}{ccccccc}
 & \vdots & & \vdots & & \vdots & \\
 & \uparrow & & \uparrow & & \uparrow & \\
A^{i-1}\otimes B^{j+1} & \longrightarrow & A^{i}\otimes B^{j+1} & \longrightarrow & A^{i+1}\otimes B^{j+1} & \longrightarrow & \dots \\
 & \uparrow & & \uparrow & & \uparrow & \\
A^{i-1}\otimes B^{j} & \longrightarrow & A^{i}\otimes B^{j} & \longrightarrow & A^{i+1}\otimes B^{j} & \longrightarrow & \dots \\
 & \uparrow & & \uparrow & & \uparrow & \\
A^{i-1}\otimes B^{j-1} & \longrightarrow & A^{i}\otimes B^{j-1} & \longrightarrow & A^{i+1}\otimes B^{j-1} & \longrightarrow & \dots
\end{array}$$

with horizontal and vertical differentials

$$\partial^h_{i,j} = \partial^i_A \otimes \mathrm{id}_{B^j}, \qquad \partial^v_{i,j} = (-1)^i \,\mathrm{id}_{A^i} \otimes \partial^j_B.$$

With this sign, the diagram anti-commutes, i.e. $\partial^h_{i,j+1} \circ \partial^v_{i,j} = -\partial^v_{i+1,j} \circ \partial^h_{i,j}$. Recall that the total complex $T(A^\bullet \otimes B^\bullet)$ in this case is defined as

$$T^n(A^\bullet \otimes B^\bullet) = \bigoplus_{i+j=n} A^i \otimes B^j,$$

with differential given on each component by the sum of the horizontal and vertical differentials:

$$\partial^n = \sum_{i+j=n} \partial^h_{i,j} + \partial^v_{i,j}.$$

Because of the anti-commutativity, we have $\partial^{n+1} \circ \partial^n = 0$, which shows that this is a complex.

Definition 3.16. We denote by $(A \otimes B)^\bullet$ the total complex $T(A^\bullet \otimes B^\bullet)$, and call it the *tensor product* of the two complexes.

Assume moreover that A, B are abelian groups, and consider the complexes $\mathrm{Hom}(A^\bullet, A)$ and $\mathrm{Hom}(B^\bullet, B)$, with differentials given by pre-composition with ∂_A, ∂_B. We want to define a product

$$H^i(\mathrm{Hom}(A^\bullet, A)) \times H^j(\mathrm{Hom}(B^\bullet, B)) \to H^{i+j}(\mathrm{Hom}((A \otimes B)^\bullet, A \otimes B)).$$

If $\alpha \in \mathrm{Hom}(A^i, A)$ and $\beta \in \mathrm{Hom}(B^j, B)$, then $\alpha \otimes \beta : A^{-i} \otimes B^{-j} \to A \otimes B$ and we get a map

$$\mathrm{Hom}(A^{-i} \otimes B^{-j}, A \otimes B) \to \mathrm{Hom}\left(\bigoplus_{k+l=i+j} A^{-k} \otimes B^{-l}, A \otimes B \right) = \mathrm{Hom}(T^{i+j}(A^\bullet \otimes B^\bullet), A \otimes B),$$

and one easily checks that this is compatible with differentials (sends cocycles to cocycles and coboundaries to coboundaries). Therefore, it induces such a product on cohomology, which is the desired cup-product.

Lemma 3.17. Let $P_\bullet \to \mathbf{Z}$ be a projective resolution of $\mathbf{Z}$ as a trivial G-module, and recall the homological indexing $P_i = P^{-i}$. Then, $(P \otimes P)_\bullet$ is a projective resolution of $\mathbf{Z}$ as a trivial $(G \times G)$-module, where the action is the diagonal one.

Applying the previous construction with $A^\bullet = B^\bullet = P_\bullet$, we get a product

$$H^i(\mathrm{Hom}_G(P_\bullet, A)) \times H^j(\mathrm{Hom}_G(P_\bullet, B)) \to H^{i+j}(\mathrm{Hom}_{G\times G}((P \otimes P)_\bullet, A \otimes B)),$$

and composing with the map $H^{i+j}(\mathrm{Hom}_{G\times G}((P\otimes P)_\bullet, A\otimes B)) \to H^{i+j}(\mathrm{Hom}_G((P\otimes P)_\bullet, A\otimes B))$ induced by the diagonal embedding $G \hookrightarrow G \times G$, we get a cup-product

$$H^i(G, A) \times H^j(G, B) \to H^{i+j}(G, A \otimes B).$$

This is compatible with restriction, inflation and corestriction maps: if $H \leq G$ is a subgroup, then

- For $a \in H^i(G, A)$ and $b \in H^j(G, B)$, we have $\mathrm{Res}(a \cup b) = \mathrm{Res}(a) \cup \mathrm{Res}(b) \in H^{i+j}(H, A \otimes B)$.
- If $H \triangleleft G$ is normal, then for $a \in H^i(G/H, A^H)$ and $b \in H^j(G/H, B^H)$, we have $\mathrm{Inf}(a \cup b) = \mathrm{Inf}(a) \cup \mathrm{Inf}(b) \in H^{i+j}(G, A \otimes B)$.
- If $[G : H] < \infty$, then for $a \in H^i(H, A)$ and $b \in H^j(G, B)$, we have

  $$\mathrm{Cor}(a \cup \mathrm{Res}(b)) = \mathrm{Cor}(a) \cup b \in H^{i+j}(G, A \otimes B)$$

  which is called the *projection formula*.

3.6 The Brauer group

Let now k be a field with a fixed separable closure k^s, and recall we defined the Brauer group of k as the second cohomology group of the absolute Galois group G_k with coefficients in $(k^s)^\times$.

Definition 3.18. The *relative Brauer group* of a separable algebraic extension $K|k$ is

$$\mathrm{Br}(K|k) \coloneqq \ker(\mathrm{Br}(k) \xrightarrow{\mathrm{Res}} \mathrm{Br}(K)).$$

Via the interpretation of the Brauer group in terms of central division algebras, this is the subgroup of $\mathrm{Br}(k)$ of those classes which split over K, i.e. which correspond to some D such that $D \otimes_k K \cong M_n(K)$ with $n = \sqrt{\dim_k D}$.

Lemma 3.19.

1. If $K|k$ is a Galois extension with group G, then $\mathrm{Br}(K|k) \cong H^2(G, K^\times)$.
2. $\mathrm{Br}(k)$ is the union of the subgroups $\mathrm{Br}(K|k)$ where K ranges over all finite Galois extensions of k.
3. If moreover G is finite cyclic, then $\mathrm{Br}(K|k) \cong k^\times / N_{K|k}(K^\times)$, where $N_{K|k}$ is the norm map.

Proof.

1. Since $H^1(G, K^\times) = 0$ by Hilbert's Theorem 90, the inflation-restriction exact sequence tells us
$$0 \to H^2(G, K^\times) \to H^2(k, (k^s)^\times) \xrightarrow{\text{Res}} H^2(K, (k^s)^\times),$$
which gives the desired isomorphism.

2. By the previous point, we have $\text{Br}(K|k) \cong H^2(G_k/G_K, K^\times)$, and the right-hand side is the direct limit of the $H^2(G_k/G_K, K^\times)$; since the transition maps are injective, this is just the union.

3. If G is cyclic and A is a G-module, then $H^2(G, A) \cong A^G/NA$ by the periodicity of cohomology. However, $A^G = k^\times$ and $NA = N_{K|k}(K^\times)$, which concludes.

□

Example 3.20.

- If $k = k^s$, then $\text{Br}(k) = 0$.
- If $k = \mathbf{R}$, then we have a non-trivial group
$$\text{Br}(\mathbf{R}) = \text{Br}(\mathbf{C}|\mathbf{R}) = H^2(\text{Gal}(\mathbf{C}|\mathbf{R}), \mathbf{C}^\times) \cong \mathbf{R}^\times/N_{\mathbf{C}|\mathbf{R}}(\mathbf{C}^\times) = \mathbf{R}^\times/\mathbf{R}_{>0} \cong \mathbf{Z}/2\mathbf{Z},$$
which on the algebras side is generated by the class of the quaternions $\mathbb{H}$.
- If $k = \mathbb{F}_q$ is a finite field, then $\text{Br}(k) = 0$. By the second part of the Lemma, it's enough to show that $\text{Br}(\mathbb{F}_{q^r}|\mathbb{F}_q) = 0$ for all $r \geq 1$. On the other hand, we know that $\text{Gal}(\mathbb{F}_{q^r}|\mathbb{F}_q)$ is generated by the Frobenius automorphism $a \mapsto a^q$. Hence,
$$N_{\mathbf{F}_{q^r}|\mathbf{F}_q}(a) = a^{1+q+\ldots+q^{r-1}} = a^{\frac{q^r-1}{q-1}}$$
and the multiplicative group of a finite field is cyclic, so if ω is a generator of $\mathbf{F}_{q^r}^\times$ then $N(\omega)$ is a generator of $\mathbf{F}_q^\times$. This implies that the norm map is surjective, and so we're done by the third part of the Lemma.

4 Local fields

Definition 4.1. A *discretely valued field* is a field K equipped with a discrete valuation map
$$v : K^\times \to \mathbf{Z} \cup \{\infty\}$$
such that

- $v(x) = \infty$ if and only if $x = 0$;
- $v(xy) = v(x) + v(y)$,
- $v(x + y) \geq \min\{v(x), v(y)\}$.

Such a field is called *complete* if it is complete as a normed space with respect to the norm $||x||_v = e^{-v(x)}$ (where e can be replaced by any $c > 1$). The *ring of integers* of K is the subring

$$\mathcal{O}_K = \{x \in K : v(x) \geq 0\} \subset K,$$

whose maximal ideal is $M_K = \{x \in K : v(x) > 0\}$; this is a principal ideal generated by any element π with $v(\pi) = 1$, which makes $\mathcal{O}_K$ a discrete valuation ring. The quotient $k(v) = \mathcal{O}_K/M_K$ is called the *residue field* of K.

Fact 4.2. The field K is complete if and only if $\mathcal{O}_K$ is complete with respect to the M_K-adic topology, i.e.

$$\mathcal{O}_K \cong \varprojlim_i \mathcal{O}_K/M_K^i.$$

Example 4.3.

- $K = \mathbf{Q}_p = \mathrm{Frac}(\mathbf{Z}_p)$, the field of p-adic numbers.
- $K = k((t)) = \mathrm{Frac}(k[[t]])$, the field of Laurent series over k.

Facts 4.4.

1. Let K be a complete discretely valued field with residue field k. Then, for any finite separable extension $L|K$ there is a unique extension of the valuation v to L, given by $w := \frac{1}{[L:K]} v \circ N_{L|K}$.
2. Moreover, L is complete with respect to this valuation, and $\mathcal{O}_L$ is the integral closure of $\mathcal{O}_K$ in L, and it is thus a finitely generated $\mathcal{O}_K$-module.

Definition 4.5. In the above situation, the *ramification index* associated with $L|K$ is $e := [w(L^\times) : v(K^\times)]$, and the *residue class degree* is $f := [k(w) : k(v)]$.

3. The following fundamental formula holds:

$$[L : K] = e \cdot f.$$

Definition 4.6. An extension $L|K$ is called *unramified* if $e = 1$, and *totally ramified* if $f = 1$.

4. If $L|K$ is finite separable, then there exists a unique intermediate extension $L \underset{e}{\supset} M \underset{f}{\supset} K$ such that $M|K$ is unramified and $L|M$ is totally ramified.
5. If $L|K$ is finite unramified and $L'|L$ is finite unramified, then $L'|K$ is finite unramified. Moreover, if $L|K$ is finite unramified and $K'|K$ is finite separable, then the composite $LK'|K'$ is finite unramified.

Definition 4.7. The maximal unramified extension of K is

$$K_{\mathrm{nr}} \coloneqq \bigcup_L \{L \subset \overline{K} : L|K \text{ finite unramified}\},$$

which has a discrete valuation extending that of K, but is **not** complete in general. Its residue field is the separable closure of $k(v)$.

As a special case, if $k(v)$ is finite then any $L|K$ finite unramified is Galois with cyclic Galois group; in particular, any divisor n of $[L : K]$ corresponds to a unique unramified extension $L \supset K_n \supset K$ of degree n over K. Therefore, in this case there exists a *unique* unramified Galois extension of degree n inside a fixed separable closure $\overline{K}$.

Let now K be a complete discretely valued field, and denote by U_K the multiplicative group of units in $\mathcal{O}_K$. We consider the following exact sequence (where we are assuming v to be surjective):

$$1 \to U_K \to K^\times \xrightarrow{v} \mathbf{Z} \to 0. \tag{4.1}$$

The surjection $U_K \to k(v)^\times$ defined by reduction modulo M_K lets us define

Definition 4.8. The group of *principal units* (or *Einsenheitengruppe*) of K is

$$U_K^1 \coloneqq \ker(U_K \to k(v)^\times),$$

and similarly for $i \geq 1$ we define

$$U_K \supset U_K^i \coloneqq \{1 + a \in K^\times : v(a) \geq i\}.$$

Since $\cap_{i \geq 1} U_K^i = \{1\}$, we have a filtration of U_K by the U_K^i. Moreover, the map $1 + a \mapsto a \mod M_K^{i+1}$ induces an isomorphism

$$U_K^i / U_K^{i+1} \cong M_K^i / M_K^{i+1} \cong \pi^i \mathcal{O}_K / \pi^{i+1} \mathcal{O}_K \cong k(v)$$

of additive groups. In particular, the subsequent quotients of the filtration are isomorphic to the additive group of the residue field, except for the first one which is $k(v)^\times$ by definition.

In the case where $K|\mathbf{Q}_p$ is finite, the exact sequence (4.1) splits, and if $q = p^{[}K : \mathbf{Q}_p]$ we have an isomorphism

$$K^\times \cong \mathbf{Z} \times \mu_{q-1} \times U_K^1.$$

More is true: U_K^1 is the direct product of μ_{p^r} and $\mathbf{Z}_p^d$, where $d = [K : \mathbf{Q}_p]$ and r is the largest integer such that $\mu_{p^r} \subset K$. In this case, we see that $K^\times / K^{\times n}$ is **finite** for all n.

Our goal now is to show Hasse's Theorem 3.14 in the following generality: if K is a completely discretely valued field with finite residue field (e.g. a finite extension of $\mathbf{Q}_p$), then $\mathrm{Br}(K) \cong \mathbf{Q}/\mathbf{Z}$ (the finite residue field assumption can actually be relaxed). We'll first show that $\mathrm{Br}(K_{\mathrm{nr}}|K) \cong \mathbf{Q}/\mathbf{Z}$, and then prove that $\mathrm{Br}(K_{\mathrm{nr}}|K) \cong \mathrm{Br}(K)$.

Lemma 4.9. Let $L|K$ be a finite Galois unramified extension with Galois group G. Then, there exists a split exact sequence

$$1 \to H^2(G, U_L) \to H^2(G, L^\times) \to \mathrm{Hom}(G, \mathbf{Q}/\mathbf{Z}) \to 1.$$

Proof. The exact sequence

$$1 \to U_L \to L^\times \xrightarrow{v} \mathbf{Z} \to 0$$

is split by sending $1 \in \mathbf{Z}$ to any uniformizer $\pi \in L^\times$, and this is G-equivariant since $L|K$ is unramified. Therefore, we get a split exact sequence

$$0 \to H^2(G, U_L) \to H^2(G, L^\times) \to H^2(G, \mathbf{Z}) \to 0$$

either by using the induced retraction and noticing that $H^1(G, \mathbf{Z}) = \mathrm{Hom}(G, \mathbf{Z}) = 0$, or directly from the fact that group cohomology is additive in the coefficients. Finally, we see that $H^2(G, \mathbf{Z}) \cong \mathrm{Hom}(G, \mathbf{Q}/\mathbf{Z})$ from the long exact sequence associated with

$$0 \to \mathbf{Z} \to \mathbf{Q} \to \mathbf{Q}/\mathbf{Z} \to 0$$

and the fact that $H^i(G, \mathbf{Q}) = 0$ for $i > 0$. □

Proposition 4.10. Let $L|K$ and G be as above, and let $\lambda|\kappa$ be the corresponding residue field extension. Then, the reduction map $U_L \to \lambda^\times$ induces an isomorphism

$$H^2(G, U_L) \cong H^2(G, \lambda^\times) = \mathrm{Br}(\lambda|\kappa).$$

Corollary 4.11. We have a split exact sequence

$$0 \to \mathrm{Br}(\lambda|\kappa) \to \mathrm{Br}(L|K) \to \mathrm{Hom}(G, \mathbf{Q}/\mathbf{Z}) \to 0.$$

In particular, when κ is finite and $[L : K] = n$ then G must be cyclic of order n, the LHS is trivial and thus $\mathrm{Br}(L|K) \cong \mathrm{Hom}(G, \mathbf{Q}/\mathbf{Z}) \cong \mathbf{Z}/n\mathbf{Z}$.

Lemma 4.12. Let G be a finite group and (A_j, φ_j) (where $\varphi_j : A_{j+1} \to A_j$) be an inverse system of G-module indexed by $\mathbf{Z}_+$, and suppose that $H^i(G, A_1) = 0$ and $H^i(G, \ker(\varphi_j)) = 0$ for some i. Then,

$$H^i(G, \varprojlim_j A_j) = 0.$$

(left as an exercise.)

Lemma 4.13 (Additive form of Hilbert 90)**.** If k is a field and $L|k$ is a finite Galois extension with group G, then $H^i(G, L^+) = 0$ for all $i > 0$.

Corollary 4.14. Passing to the limit, we get $H^i(k,(k^s)^+)=0$ for all $i>0$.

Proof of Lemma. In characteristic 0, we know this because L^+ is a $\mathbf{Q}$-vector space. In general, by the normal basis theorem there is some $a\in L$ such that $\{\sigma(a):\sigma\in G\}$ is a k-basis of L, so

$$L^+\cong k^+\otimes_{\mathbf{Z}}\mathbf{Z}[G]$$

as G-modules.

Since G is finite, this is non-canonically isomorphic to the co-induced module

$$\mathrm{Hom}(\mathbf{Z}[G],k^+)=M^G(k^+),$$

and the latter has trivial cohomology by Shapiro's lemma. □

Proof of Proposition 4.10. Notice that, since $\cap_{j=1}^{\infty}U_L^j=\{1\}$, $U_L^{(1)}$ is the inverse limit of the quotients U_L^1/U_L^j. Therefore, we have exact sequences

$$0\to U_L^{(1)}\to U_L\to\lambda^\times\to 0, \tag{4.2}$$

$$0\to U_L^{j+1}\to U_L^j\to\lambda^+\to 0\quad(j\geq 1). \tag{4.3}$$

By the second lemma, we know that $H^i(G,\lambda^+)=0$ for $i>0$, so that $H^i(G,U_L^j/U_L^{j+1})=0$ for $i>0$. Therefore, by the first lemma we get $H^i(G,U_L^{(1)})=0$ for $i>0$, since

$$U_L^{(j)}/U_L^{(j+1)}=\ker(U_L^{(1)}/U_L^{(j+1)}\to U_L^{(1)}/U_L^{(j)}).$$

Finally, the long exact sequence associated with (4.2) gives

$$\cancel{H^2(G,U_L^{(1)})}\to H^2(G,U_L)\xrightarrow{\sim}H^2(G,\lambda^\times)\to\cancel{H^3(G,U_L^{(1)})},$$

□

In summary, by passing to the limit we get a split exact sequence

$$0\to\mathrm{Br}(k(v)^s|k(v))\to\mathrm{Br}(K_{\mathrm{nr}}|K)\to\mathrm{Hom}(G_k,\mathbf{Q}/\mathbf{Z})\to 0,$$

The next step is to show that $\mathrm{Br}(K_{\mathrm{nr}}|K)$ is the full Brauer group of K. This is true for any complete discretely valued field, but we only give the proof in the case of finite residue field.

Lemma 4.15. If K is a complete discretely valued field and $L|K$ is a finite Galois extension with group G, then U_L contains a G-submodule $V_L^1\subset U_L$ of finite index such that $H^i(G,V_L^1)=0$ for all $i>0$.

Proof. By the normal basis theorem we have a K-basis $\{\sigma(a):\sigma\in G\}$ of L for some $a\in L$. Let π be a uniformizer of K (that is, $v(\pi)=1$), and let r be large enough so that $pi^r\sigma(a)\in\mathcal{O}_L$ for all $\sigma\in G$.

Let M be the $\mathcal{O}_K$-submodule of $\mathcal{O}_L$ generated by the $\pi^r\sigma(a)$. This is a G-submodule of $\mathcal{O}_L$, and recall that $\mathcal{O}_L$ is a free $\mathcal{O}_K$-module of rank $[L:K]$, so M is as well. But then $\mathcal{O}_L/M$ must be

finite, because it is a finitely generated torsion $\mathcal{O}_K$-module. Now, fix m such that $\pi^m \mathcal{O}_L \subset M$, and let

$$V_L^j \coloneqq 1 + \pi^j M \subset U_L^j$$

for $j \geq 1$. □

Corollary 4.16. The group $H^i(G, U_L)$ is finite for all $i > 0$.

Proof. □

Proposition 4.17. If $L|K$ are as above and $G \cong \mathbf{Z}/n\mathbf{Z}$, then $|\mathrm{Br}(L|K)| = n$.

Lemma 4.18. If G is finite cyclic and A is a *finite* G-module, then $|H^2(G, A)| = |H^1(G, A)|$.

Proof. Consider the following exact sequences of finite abelian groups, where $G = \langle \sigma \rangle$:

$$0 \to A^G \to A \xrightarrow{\sigma - 1} A \to (\sigma - 1)A \to 0,$$
$$0 \to {}_N A \to A \xrightarrow{N} NA \to 0.$$

Then $H^1(G, A) = {}_N A/(\sigma - 1)A$ and $H^2(G, A) = A^G/NA$, so the result follows from the fact that $|A| = |A^G| \cdot |(\sigma - 1)A| = |nA| \cdot |NA|$. □

Proof of Proposition 4.17. We need to show that $H^2(G, L^\times)$ is finite of order n. The long exact sequence associated with $1 \to U_L \to L^\times \xrightarrow{v} \mathbf{Z} \to 0$ looks like

$$\cancel{H^1(G, \mathbf{Z})} \to H^2(G, U_L) \to H^2(G, L^\times) \to H^2(G, \mathbf{Z}) \cong \mathrm{Hom}(G, \mathbf{Q}/\mathbf{Z}) \to H^3(G, U_L) \to H^3(G, L^\times) \cong \cancel{H^1(G, L^\times)}$$

where the last term is trivial by Hilbert 90. By Lemma 4.15, we have

$$H^i(G, U_L) \cong H^i(G, U_L/V_L^1) \quad (i > 0),$$

and by the Corollary the RHS is finite. Therefore, $H^2(G, L^\times)$ has a map into $H^2(G, \mathbf{Z}) = \mathrm{Hom}(G, \mathbf{Q}/\mathbf{Z}) \cong \mathbf{Z}/n\mathbf{Z}$ with finite kernel and cokernel of the same order by the previous lemma (since $H^3(G, U_L) = H^1(G, U_L)$), so it must be finite of order n as well. □

We need the following compatibility result:

Proposition 4.19. If $L|K$ is a finite separable extension, then the following diagram commutes:

$$\begin{array}{ccc} \mathrm{Br}(K_{\mathrm{nr}}|K) & \xrightarrow{\cong} & \mathbf{Q}/\mathbf{Z} \\ \Big\downarrow{\scriptstyle \mathrm{Res}} & & \Big\downarrow{\scriptstyle \cdot [L:K]} \\ \mathrm{Br}(L_{\mathrm{nr}}|L) & \xrightarrow{\cong} & \mathbf{Q}/\mathbf{Z} \end{array}$$

Proof. Recal that if $L \supset M \supset K$ is the maximal unramified subextension of $L|K$, it's enough to prove the Proposition fo $L = M$ or $M = K$.

In the first case, $L_{\mathrm{nr}} = K_{\mathrm{nr}}$ and if κ, λ are the residue fields of K, L respectively, then we have the long exact sequence

$$\mathrm{Br}(K_{\mathrm{nr}}|K) = H^2(\mathrm{Gal}(K_{\mathrm{nr}}|K), K_{\mathrm{nr}}^\times) \xrightarrow{v} H^2(\mathrm{Gal}(K_{\mathrm{nr}}|K), \mathbf{Z}) \cong H^2(\mathrm{Gal}(\overline{\kappa}|\kappa), \mathbf{Z}) \cong H^1(\kappa, \mathbf{Q}/\mathbf{Z})$$

which maps via Res to the corresponding long exact sequence for L:

$$\mathrm{Br}(L_{\mathrm{nr}}|L) = H^2(\mathrm{Gal}(L_{\mathrm{nr}}|L), L_{\mathrm{nr}}^\times) \xrightarrow{v} H^2(\mathrm{Gal}(L_{\mathrm{nr}}|L), \mathbf{Z}) \cong H^2(\mathrm{Gal}(\overline{\lambda}|\lambda), \mathbf{Z}) \cong H^1(\lambda, \mathbf{Q}/\mathbf{Z}).$$

In practice, the rightmost terms are continuous homomorphisms from the absolute Galois groups of the residue fields to $\mathbf{Q}/\mathbf{Z}$ (which is just $\mathbf{Q}/\mathbf{Z}$), and the map which makes the diagram commute is multiplication by $[L : K] = [\lambda : \kappa]$.

In the latter case, if $L|K$ is totally ramified than $\kappa = \lambda$ and L is linearly disjoint from K_{nr} over K, so that $L_{\mathrm{nr}} = LK_{\mathrm{nr}}$. Therefore, we have an analogous diagram since $v_L \circ \mathrm{Res} = [L : K] \cdot \mathrm{Res} \circ v_K$ by the definition of the valuation on L. Thus, the rightmost vertical map induces multiplication by $[L : K]$ on $H^1(\kappa, \mathbf{Q}/\mathbf{Z})$, which concludes the proof. □

Finally, we deduce:

Proposition 4.20. If K is a complete discrete valuation field with finite residue field, then $\mathrm{Br}(K_{\mathrm{nr}}|K) \cong \mathrm{Br}(K)$.

Proof. Let $K_n|K$ be the unique unramified extension of degree n. Let $L|K$ be any finite Galois extension of degree n. We prove that $\mathrm{Br}(L|K) \cong \mathrm{Br}(K_n|K)$ as subgroups of $\mathrm{Br}(K)$, which implies the result by taking the direct limit over all n.

- $\mathrm{Br}(K_n|K) \subset \mathrm{Br}(L|K)$: we have a diagram

  We know that if $L|K$ is cyclic, then $|\mathrm{Br}(L|K)| = n$. Otherwise, we proceed by cases:

- If $\mathrm{Gal}(L|K)$ is a p-group of order $n = p^r$, then it is solvable and so there is a chain of subfields
  $$K = L_0 \subset L_1 \subset \ldots \subset L_r = L$$
  such that $L_i|K$ is Galois for all i and $\mathrm{Gal}(L_i|L_{i-1})$ is cyclic of order p. By the inflation-restriction sequence, for all i we have exact sequences
  $$0 \to \mathrm{Br}(L_i|L_{i-1}) \to \mathrm{Br}(L_i|K) \to \mathrm{Br}(L_{i-1}|K),$$
  and by induction we see that the LHS has order p, the RHS has order dividing p^{i-1}, and therefore $|\mathrm{Br}(L_i|K)|$ divides p^i. For $i = r$, this means that $|\mathrm{Br}(L|K)|$ divides $p^r = n$, but since $\mathrm{Br}(K_n|K) \subset \mathrm{Br}(L|K)$ and $|\mathrm{Br}(K_n|K)| = n$, equality must hold.

- In the general case, $n = \prod_{i=1}^m p_i^{r_i}$ and we can choose Sylow p-subgroups P_i of $\mathrm{Gal}(L|K)$. Therefore,
  $$\mathrm{Br}(L|K) \cong \bigoplus_{i=1}^m \mathrm{Br}(L|K)\{p_i\}$$

where $\{p_i\}$ indicates the p_i-primary torsion, and the composite

$$\mathrm{Br}(L|K)\{p_i\} = H^2(G, L^\times)\{p_i\} \xrightarrow{\mathrm{Res}} H^2(P_i, L^\times) \xrightarrow{\mathrm{Cor}} H^2(G, L^\times)$$

is injective since it coincides with multiplication by $[G : P_i]$,which is coprime to p_i. Therefore, Res is injective on the p_i-primary torsion, and thus

$$\mathrm{Br}(L|K) \hookrightarrow \bigoplus_{i=1}^{m} H^2(P_i, L^\times) = \bigoplus_{i=1}^{m} \mathrm{Br}(L|K_i),$$

where K_i is the fixed field of P_i. By the case of p-groups, each term has order dividing $p_i^{r_i}$, and we conclude that $|\mathrm{Br}(L|K)|$ divides $\prod_{i=1}^m p_i^{r_i} = n$ also in this case.

□

We can finally conclude the proof of Hasse's theorem:

Theorem 4.21 (Hasse)**.** If K is a complete discretely valued field with finite residue field, then there exists a canonical isomorphism $\mathrm{inv} : \mathrm{Br}(K) \xrightarrow{\sim} \mathbf{Q}/\mathbf{Z}$ called the *Hasse invariant*. Moreover, the following diagrams commute for $L|K$ finite separable:

$$\begin{array}{ccc} \mathrm{Br}(K) & \xrightarrow{\mathrm{inv}} & \mathbf{Q}/\mathbf{Z} \\ {\scriptstyle \mathrm{Res}}\downarrow & & \downarrow{\scriptstyle \cdot [L:K]} \\ \mathrm{Br}(L) & \xrightarrow{\mathrm{inv}} & \mathbf{Q}/\mathbf{Z} \end{array}$$

and

Proof. The only thing left to prove is the commutativity of the second diagram. Observe that the map $\mathrm{Br}(K) \xrightarrow{\mathrm{Res}} \mathrm{Br}(L)$ is surjective because $\cdot[L : K]$ is surjective on the divisible group $\mathbf{Q}/\mathbf{Z}$. So, if $\alpha \in \mathrm{Br}(L)$ then there exists $\beta \in \mathrm{Br}(K)$ such that $\mathrm{Res}(\beta) = \alpha$. Then, we have

$$\mathrm{Cor}(\alpha) = \mathrm{Cor}(\mathrm{Res}(\beta)) = [L : K] \cdot \beta,$$

and thus $\mathrm{inv}(\alpha) = \mathrm{inv}([L : K] \cdot \beta) = [L : K] \cdot \mathrm{inv}(\beta) = \mathrm{inv}(\mathrm{Cor}(\alpha))$ by the first diagram, which concludes the proof. □

Corollary 4.22. We have $H^2(K, \mu_n) \cong \mathbf{Z}/n\mathbf{Z}$ for n prime to $\mathrm{char}(K)$.

Proof. Given our computation of the Brauer group, this follows from the long exact sequence associated with the Kummer exact sequence

$$1 \to \mu_n \to K^s \xrightarrow{(\cdot)^n} K^s \to 1$$

and the fact that $H^1(K, (K^s)^\times) = 0$ by Hilbert 90, from which

$$H^2(K, \mu_n) \cong H^2(K, (K^s)^\times)[n] = \mathrm{Br}(K)[n] \cong \frac{1}{n}\mathbf{Z}/\mathbf{Z} \cong \mathbf{Z}/n\mathbf{Z}.$$

□

Construction 4.23. We construct the Hilbert symbol for $(n, \mathrm{char}(K)) = 1$ as follows. For $\chi \in H^1(K, \mathbf{Z}/n\mathbf{Z})$ and $b \in K^\times/K^{\times n} \cong H^1(K, \mu_n)$, we have a cup product

$$\chi \cup b \in H^2(K, \mu_n) \cong \mathbf{Z}/n\mathbf{Z},$$

which we denote by (χ, b).

In particular, in the case where $K \supset \mu_n$, we have an isomorphism $H^1(K, \mathbf{Z}/n\mathbf{Z}) \cong H^1(K, \mu_n)$ which depends on the choice of a primitive n-th root of unity $\omega \in \mu_n$, so we get a pairing

$$(\cdot, \cdot) : K^\times/K^{\times n} \times K^\times/K^{\times n} \to \mathbf{Z}/n\mathbf{Z}$$

sending (a, b) to (χ_a, b), where χ_a is the character corresponding to a. Moreover, by using the same fixed isomrphism $\mathbf{Z}/n\mathbf{Z} \cong \mu_n$ we get a pairing

$$(\cdot, \cdot) : K^\times/K^{\times n} \times K^\times/K^{\times n} \to \mu_n$$

which does not depend on the choice of ω.

The following results helps us verify that this really is a generalization of the classical Hilbert symbol, even for $n = 2$:

Proposition 4.24. Suppose that $\chi \in H^1(K\mathbf{Z}/n\mathbf{Z})$ corresponds to a *surjective* character $\mathrm{Gal}(K_s|K) \twoheadrightarrow \mathbf{Z}/n\mathbf{Z}$, and let L be the fixed field of its kernel. Then, if $b \in K^\times$

$$(\chi, b) = 0 \iff b \in N_{L|K}(L^\times).$$

Suppose $n = 2$ (so that $\mu_2 = \{\pm 1\} \subset K$). Then, L is a quadratic extension $K(\sqrt{a})$ for some $a \in K^\times$, and the above proposition says that $(a, b) = 0$ if and only if b is a norm from $K(\sqrt{a})$, that is there are some $x, y \in K$ such that

$$N_{K(\sqrt{a})|K}(x + y\sqrt{a}) = x^2 + ay^2 = b \iff x^2 + ay^2 - bz^2 \text{ has a nontrivial zero;}$$

which was precisely our original definition of the Hilbert symbol.

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